

RAOUL ISLAND, New Zealand

WMO: 939940

Lat: 29.245S

Lon: 177.929W

Elev: 50

StdP: 100.73

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.3	12.1	5.8	5.8	14.8	6.7	6.1	14.8	9.5	17.4	8.6	17.4	1.9	220	0.322

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.4	27.0	23.6	26.2	22.8	25.5	22.2	24.5	26.1	23.9	25.3	23.2	24.8	3.0	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.0	25.5	23.4	18.2	25.0	22.7	17.5	24.5	74.6	26.2	72.0	25.4	69.6	24.8	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.1	8.0	7.2	9.8	28.6	0.8	1.0	9.3	29.4	8.8	30.0	8.3	30.6	7.8	31.4
			8.3	25.3	0.7	0.8	7.8	25.9	7.4	26.4	7.0	26.8	6.5	27.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.4	22.5	23.1	22.7	21.1	19.2	17.5	16.5	16.0	16.7	17.6	19.1	21.0
	DBStd	2.85	1.49	1.27	1.32	1.55	1.43	1.63	1.54	1.43	1.37	1.35	1.52	1.45
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	265	0	0	0	1	8	36	59	72	51	30	9	1
	CDD10.0	3427	387	366	393	333	284	223	202	187	202	237	273	340
	CDD18.3	652	128	133	134	83	33	9	3	1	3	8	32	82
	CDH23.3	1197	316	383	303	86	3	0	0	0	0	0	9	97
	CDH26.7	37	14	14	8	1	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	3.3	3.6	3.7	3.6	3.5	3.3	3.3	3.4	3.2	3.2	3.0	2.9	3.1
(15) Precipitation	PrecAvg	1468	115	140	144	122	120	145	143	129	111	82	92	100
	PrecMax	2143	464	365	404	539	254	376	321	340	258	234	318	377
	PrecMin	865	12	11	28	20	28	45	48	9	37	15	2	8
	PrecStd	326	97	92	94	86	55	72	59	59	55	45	66	76
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.2	28.1	27.7	26.2	23.8	22.4	21.2	20.7	21.4	22.5	24.4	26.3
		MCWB	24.8	24.5	24.6	23.1	20.1	20.0	18.3	18.0	18.3	19.0	21.1	22.8
	2%	DB	26.8	27.0	26.5	25.1	22.8	21.4	20.4	19.9	20.5	21.6	23.4	25.2
		MCWB	23.4	23.2	23.5	22.0	19.4	19.0	17.6	16.9	17.4	18.3	20.1	21.8
	5%	DB	25.7	26.2	25.6	24.3	22.2	20.7	19.7	19.2	19.9	21.0	22.7	24.4
		MCWB	22.3	22.6	22.7	21.3	18.9	18.3	17.0	16.2	16.9	17.6	19.4	21.1
	10%	DB	25.0	25.4	24.9	23.6	21.5	20.2	19.1	18.7	19.3	20.3	22.0	23.7
		MCWB	21.7	22.1	21.9	20.7	18.4	17.6	16.5	15.7	16.5	17.1	18.8	20.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.4	25.1	25.2	23.9	21.5	21.0	19.6	19.2	19.5	20.2	22.4	23.7
		MCDB	27.0	27.2	26.7	25.2	22.8	21.7	20.5	20.1	20.4	21.5	23.5	25.3
	2%	WB	24.4	24.3	24.4	23.0	20.6	20.0	18.7	18.2	18.7	19.4	21.3	22.7
		MCDB	26.0	25.8	25.9	24.3	21.9	20.9	19.8	19.2	19.6	20.7	22.4	24.3
	5%	WB	23.5	23.7	23.7	22.2	20.0	19.1	17.8	17.4	18.0	18.7	20.5	22.0
		MCDB	25.1	25.1	25.1	23.7	21.4	20.3	19.2	18.6	19.2	20.1	21.8	23.6
	10%	WB	22.6	23.1	22.9	21.5	19.3	18.3	17.1	16.5	17.3	18.0	19.7	21.3
		MCDB	24.3	24.7	24.3	23.2	21.0	19.7	18.7	18.1	18.9	19.6	21.3	23.1
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	4.4	4.2	4.2	4.1	4.1	4.2	4.6	4.6	4.9	4.9	4.8
		MCDBR	5.3	5.4	4.8	4.8	4.6	4.4	4.5	5.0	5.1	5.5	5.4	5.5
		MCWBR	3.0	3.0	2.7	2.9	2.9	3.3	3.3	3.4	3.3	3.4	3.2	3.1
	5% WB	MCDBR	4.4	4.3	4.0	4.1	3.5	3.6	3.5	3.7	4.0	4.4	4.2	4.3
		MCWBR	2.7	2.8	2.6	2.9	2.9	3.4	3.3	3.4	3.2	3.3	3.0	2.9
(40) Clear-Sky Solar Irradiance	taub	0.362	0.362	0.356	0.353	0.337	0.333	0.328	0.329	0.345	0.342	0.342	0.342	0.354
	taud	2.606	2.627	2.642	2.632	2.651	2.632	2.637	2.628	2.561	2.566	2.599	2.605	
	Ebn at Noon	979	961	936	890	859	838	860	902	932	970	992	990	
	Edh at Noon	103	99	92	84	75	72	75	83	98	103	103	104	
(44) All-Sky Solar Radiation	RadAvg	6.95	6.31	5.19	4.15	3.19	2.73	3.02	3.81	4.97	6.18	6.86	7.22	
	RadStd	0.48	0.40	0.33	0.33	0.28	0.14	0.20	0.17	0.37	0.30	0.43	0.46	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page